CSA0716-Computer Networks

CO1-AT1

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1. File Transfer using Transport and Data Link Layers

Given:

- File size = **150 MB** = 150 × 1024 × 1024 = **157,286,400 bytes**
- Segment size = **1500 bytes**
- Data Link overhead = **40 bytes/frame**
- Bandwidth = **50 Mbps**

Step 1: Number of Segments

157,286,4001500=104,857.6\frac{157,286,400}{1500}=104,857.61500157,286,400 =104,857.6

Segments = 104,858

Since each segment becomes one frame,

Frames = 104,858

Step 2: Total Data Link Overhead

104,858×40=4,194,320 bytes104,858 \times 40 = 4,194,320 \text{ bytes}104,858×40=4,194,320 bytes

≈ **4 MB**

Step 3: Total Data Transmitted

157,286,400+4,194,320=161,480,720 bytes157,286,400+4,194,320=161,480,720\text{ bytes}157,286,400+4,194,320=161,480,720 bytes

Convert to bits:

161,480,720×8=1,291,845,760 bits161,480,720 \times 8=1,291,845,760\text{ bits}161,480,720×8=1,291,845,760 bits

Transmission time:

$$\frac{1,291,845,760}{50\times10^6}=25.84\text{ s}$$

Answer

- Segments = **104,858**
- Frames = **104,858**
- Data Link overhead = **4,194,320 bytes (≈4 MB)**
- Transmission time = **25.84 s**

Explanation

- **Transport Layer:** Segments data, performs flow control, error recovery, acknowledgements, and retransmissions.
- **Data Link Layer:** Creates frames, detects transmission errors using CRC, and ensures reliable node-to-node delivery.

2. Sliding Window Protocol

Given

- Window size = 6
- Total frames = 48
- Frame size = 1024 bytes

Number of Windows

$$\frac{48}{6}=8$$

Windows = 8

Retransmissions

1 frame lost in every window

$$8\times1=8$$

Answer

- Windows = **8**
- Retransmissions = **8**

Explanation

Flow control prevents sender overflow, matches receiver speed, reduces congestion, and improves throughput.

3. IP Fragmentation

Given

- Packet = 12,000 bytes
- MTU = 1500 bytes
- IP Header = 20 bytes

Payload per Fragment

1500−20=1480 bytes1500-20=1480\text{ bytes}1500−20=1480 bytes

Fragments

120001480=8.11\frac{12000}{1480}=8.11148012000=8.11

Fragments = **9**

Header Overhead

9×20=180 bytes9\times20=180\text{ bytes}9×20=180 bytes

Answer

- Fragments = **9**
- Header overhead = **180 bytes**

Explanation

The Network Layer fragments packets, assigns fragment offsets and identification numbers, and reassembles them at the destination.

4. Sliding Window Protocol (Same as Question 2)

Answer

- Windows = **8**
- Retransmissions = **8**

Explanation

Flow control prevents packet loss by regulating the transmission rate according to receiver capacity.

5. Session Layer Synchronization

Given

- File = 600 MB
- Checkpoint every 60 MB
- Failure after 390 MB

Checkpoints Before Failure

390 ÷ 60 = 6 checkpoints

(Checkpoints at 60,120,180,240,300,360 MB)

Retransmission

Last checkpoint = 360 MB

Failure = 390 MB

Retransmit:

390−360=30 MB390-360=30\text{ MB}390−360=30 MB

Answer

- Checkpoints = **6**
- Retransmitted data = **30 MB**

Explanation

Synchronization checkpoints allow transmission to resume from the latest checkpoint instead of restarting from the beginning.

6. Presentation Layer Compression & Encryption

Given

- Original = 900 MB
- Compression = 40%
- Encryption overhead = 5%

Compressed Size

900×0.60=540 MB900\times0.60=540\text{ MB}900×0.60=540 MB

Encrypted Size

540×1.05=567 MB540\times1.05=567\text{ MB}540×1.05=567 MB

Percentage Reduction

900−567900×100=37%\frac{900-567}{900}\times100=37\%900900−567×100=37%

Answer

- Compressed size = **540 MB**
- Encrypted size = **567 MB**
- Total reduction = **37%**

Explanation

The Presentation Layer compresses data to reduce transmission time and encrypts it to ensure confidentiality.

7. FTP File Transfer

Given

- Data = 3 GB = 3072 MB
- Throughput = 120 Mbps
- Request size = 3 MB

Requests

30723=1024\frac{3072}{3}=102433072=1024

Transmission Time

3072 MB = 24,576 Mb

24,576120=204.8 s\frac{24,576}{120}=204.8\text{ s}12024,576=204.8 s

Answer

- Requests = **1024**
- Transmission time = **204.8 seconds**

Explanation

The Application Layer provides user services such as FTP commands, authentication, error reporting, and reliable file transfer.

8. End-to-End Delay

Processing delay:

Application = 4 ms

Transport = 3 ms

Network = 5 ms

Data Link = 2 ms

Physical = 1 ms

Total Processing:

4+3+5+2+1=15 ms4+3+5+2+1=15\text{ ms}4+3+5+2+1=15 ms

Propagation = 18 ms

Transmission = 12 ms

Total Delay:

15+18+12=45 ms15+18+12=45\text{ ms}15+18+12=45 ms

Answer

Total end-to-end delay = **45 ms**

Explanation

- Application: User services
- Transport: Segmentation and reliability
- Network: Routing
- Data Link: Framing and error detection
- Physical: Bit transmission

9. Frame Overhead

Given

- Useful data = 80 MB
- Frame size = 1600 bytes
- Overhead = 80 bytes/frame

Useful payload/frame:

1600−80=1520 bytes1600-80=1520\text{ bytes}1600−80=1520 bytes

Useful data:

80 MB = 83,886,080 bytes

Frames

83,886,0801520=55,188.2\frac{83,886,080}{1520}=55,188.2152083,886,080=55,188.2

Frames = **55,189**

Total Overhead

55,189×80=4,415,120 bytes55,189\times80=4,415,120\text{ bytes}55,189×80=4,415,120 bytes

Efficiency

15201600×100=95%\frac{1520}{1600}\times100=95\%16001520×100=95%

Answer

- Frames = **55,189**
- Overhead = **4,415,120 bytes**
- Transmission efficiency = **95%**

Explanation

Protocol overhead consumes bandwidth, reducing effective throughput and increasing transmission time.

10. Healthcare Network

Given

- Original = 240 MB
- Compression = 30%
- Segment = 1400 bytes
- Frame overhead = 60 bytes

Compressed Size

240×0.70=168 MB240\times0.70=168\text{ MB}240×0.70=168 MB

Compressed bytes:

168×1024×1024=176,160,768 bytes168\times1024\times1024=176,160,768\text{ bytes}168×1024×1024=176,160,768 bytes

Segments

176,160,7681400=125,829.12\frac{176,160,768}{1400}=125,829.121400176,160,768=125,829.12

Segments = **125,830**

Protocol Overhead

125,830×60=7,549,800 bytes125,830\times60=7,549,800\text{ bytes}125,830×60=7,549,800 bytes

Final Data

176,160,768+7,549,800=183,710,568 bytes176,160,768+7,549,800=183,710,568\text{ bytes}176,160,768+7,549,800=183,710,568 bytes

≈ **175.2 MB**

Answer

- Compressed size = **168 MB**
- Segments = **125,830**
- Protocol overhead = **7,549,800 bytes**
- Final transmitted data = **183,710,568 bytes (≈175.2 MB)**

Explanation

- **Presentation Layer:** Compresses and encrypts data.
- **Transport Layer:** Segments data and provides reliable delivery.
- **Data Link Layer:** Frames data and detects errors.
- **Physical Layer:** Transmits bits over the communication medium.